Lightweight Cryptography (CRYE6135) · Session 2, Project 2

Gate Equivalents & Multi-Technology ASIC Synthesis

Hand-Counted Gate Analysis vs. Yosys Standard-Cell Synthesis in 45 nm, 130 nm & 180 nm

Group 2

1. Problem Statement · Passive UHF RFID Tags

Why Standard Block Ciphers Exceed Tag Limits

2,000 GE

Max Security Gate Budget

Only 200 to 2,000 GE available out of 1,000 to 10,000 total tag gates.

RFID Security Ceiling — **2,000 GE (1.0×)**

Maximum silicon budget for cryptographic logic.

HIGHT Block Cipher — **3,048 GE (1.5×)**

Exceeds maximum tag security budget by 1,048 GE.

Compact AES-128 — **3,400 GE (1.7×)**

Exceeds maximum tag security budget by 1,400 GE.

2. Project Scope · Candidate RTL Circuits

Four Hardware Primitives Investigated

1. PRESENT 4-Bit S-Box

16-entry non-linear substitution table.

2. 32-Bit XOR-Rotate

ARX primitive with 7-bit fixed shift.

3. 64-Bit PRESENT Round

64-bit key XOR with 16 parallel S-boxes.

4. 8-Bit AES S-Box LUT

256-entry lookup table baseline.

3. Hardware Metric · ASIC Area Normalization

Gate Equivalents (GE) & Elementary Weights

Gate Equivalent Formula

$$\mathrm{GE} = \frac{\text{Design Area } (\mu m^2)}{\text{NAND2 Area } (\mu m^2)}$$

1 GE equals the silicon area of one 2-input NAND gate in the same technology library.

Gate Type	Textbook Weight
Wire / Bit Permutation	0.00 GE
Inverter (INV)	0.67 GE
2-Input NAND2 / NOR2	1.00 GE
2-Input AND2 / OR2	1.33 GE
2-Input XOR2	2.00 GE

4. Hand-Counted Analysis · Non-Linear S-Box

4-Bit PRESENT S-Box: 31 Gates = 41.0 GE

14	8	4	4 + 1
AND2 Gates	OR2 Gates	XOR2 Gates	INV + NAND2
18.67 GE	10.67 GE	8.00 GE	3.67 GE

Verified against all 16 PRESENT S-box outputs (C, 5, 6, B, 9, 0, A, D, 3, E, F, 8, 4, 7, 1, 2) **41.0 GE**

5. Hand-Counted Analysis · Full Round & Free Wiring

PRESENT Round (784 GE) & XOR-Rotate (64 GE)

1. addRoundKey

64 × XOR2 gates

128.0 GE

→

2. sBoxLayer

16 × 41.0 GE S-boxes

656.0 GE

→

3. pLayer Wires

64-bit wire routing

0.0 GE

32-Bit XOR-Rotate: 32 XOR2 gates (64 GE) + 7-bit wire shift (0 GE) **64.0 GE**

6. Experimental Setup · Open-Source EDA Flow

Yosys + ABC Synthesis Across Three PDKs

Reference Vectors

PRESENT-80 model passes all 4 official standard test vectors.

Three Real Libraries

Nangate 45 nm, SkyWater 130 nm, GlobalFoundries 180 nm PDKs.

Equivalence Proof

Every synthesized gate netlist is formally proven equal to RTL.

7. Experimental Results · Gate Equivalents (GE)

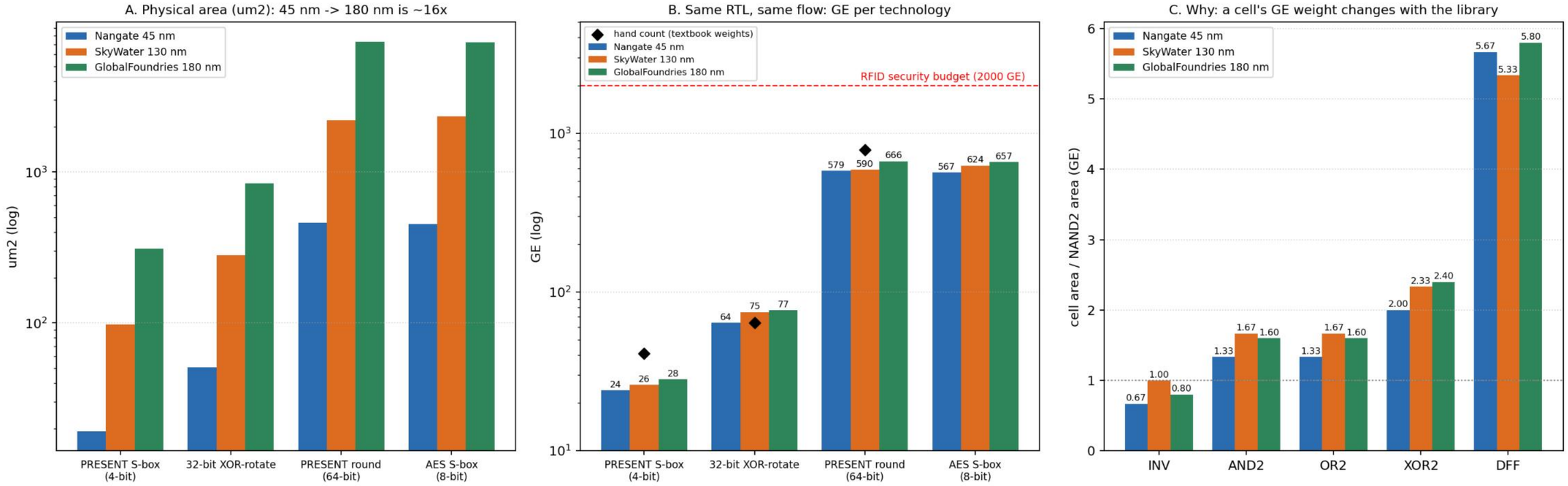
Hand Count vs. Synthesized Gate Equivalents

Hardware Design	Hand Count	Nangate 45 nm	SkyWater 130 nm	GF 180 nm
32-Bit XOR-Rotate	64.0 GE	64.0 GE	74.7 GE	76.8 GE
PRESENT S-Box (4-Bit)	41.0 GE	24.0 GE	26.0 GE	28.2 GE
PRESENT Round (64-Bit)	784.0 GE	579 GE	590 GE	666 GE
AES S-Box (8-Bit LUT)	—	567 GE	624 GE	657 GE

Combinational logic only: 64-bit state register adds ~340–370 GE (64 DFF cells).

8. Multi-Technology Plot · Area, GE & Cell Weights

45 nm vs. 130 nm vs. 180 nm Synthesis Comparison



9. Root Cause Analysis · Panel C Breakdown

Why a Cell's GE Weight Changes Per Library

INV	45 nm: 0.67 GE	130 nm: 1.00 GE	180 nm: 0.80 GE
AND2 / OR2	45 nm: 1.33 GE	130 nm: 1.67 GE	180 nm: 1.60 GE
XOR2	45 nm: 2.00 GE	130 nm: 2.33 GE	180 nm: 2.40 GE
DFF	45 nm: 5.67 GE	130 nm: 5.33 GE	180 nm: 5.80 GE

10. Core Engineering Question · Cross-Technology Limits

Why GE Numbers Must Not Be Compared Directly

1. Real Silicon Area Changes ~16×

- **16× Die Shrink:** 180 nm to 45 nm physical area.
- **Exercise 4 Proof:** 2,000 GE is 3,000 μm^2 vs 1,000 μm^2.
- **Physical Cost:** Wafer cost and capacitance depend on μm^2.

2. Same RTL Gives 15–20% Different GE

- **Non-Uniform Cells:** XOR2 costs 2.00 GE to 2.40 GE.
- **PRESENT Round Shift:** 579 GE (45nm) vs 666 GE (180nm).
- **Tool Version Effect:** ABC mapper shifts GE on same PDK.

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What We Found & What Surprised Us

41% Smaller via Synthesis

S-box drops from 41 GE by hand to 24 GE in 45 nm.

0 GE Wire Permutations

PRESENT pLayer and 7-bit rotation use zero logic gates.

24× Smaller Than AES LUT

PRESENT S-box takes 24 GE vs 567 GE for AES LUT.

Live Yosys Demo & Questions · Thank You!